Week 6 Report

HMMs on synthetic volatility regimes

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Experiment and evaluation protocol

Known volatility regimes

$$\mu = (0, 0, 0), \qquad \sigma = (1, 2, 4).$$

$A_{kk} = 0.995$; $A_{kj} = 0.0025$ for $k \neq j$.
Expected duration: 200 observations.
$\pi = (1/3, 1/3, 1/3)$.

Gaussian, Student-t ($\nu = 2.5$), trimodal, and skewed Gamma (shape 4) emissions, each centered and scaled to these moments.

Paired comparison

- 3 states; GMM has 3 components per state.
- 3 replicates, 3 starts, at most 1,000 EM updates.
- Select by clean-training likelihood; align using training labels only.
- Freeze models and alignment for every test condition.

$$X_t^{(\lambda)} = X_t + \lambda s_{\text{train}} \frac{t}{T-1} \eta_t, \quad \eta_t \sim \mathcal{N}(0,1), \quad \lambda \in \{0.10, 0.25, 0.50\}.$$

Noise SD grows from zero to λs_{train}. True labels stay fixed.

Clean test: state recovery

Means over 10 paired replicates; 20,000 training and 5,000 test observations per replicate.

	Precision (%)		Recall (%)		F1 (%)	
Family	Gaussian	GMM	Gaussian	GMM	Gaussian	GMM
Gaussian	97.94	97.76	98.11	97.95	98.01	97.84
Student-t	44.24	95.41	61.19	95.05	48.73	95.12
Trimodal	59.81	69.63	60.47	70.37	56.28	67.86
Skewed Gamma	82.85	86.19	84.27	87.18	82.06	86.40

Clean test: posterior quality and density fit

Means over 10 paired replicates; 20,000 training and 5,000 test observations per replicate.

	State log loss		Posterior Confidence (%)		Emission KL	
Family	Gaussian	GMM	Gaussian	GMM	Gaussian	GMM
Gaussian	0.050	0.051	97.27	97.25	0.0001	0.0002
Student-t	2.125	0.114	55.00	94.44	1.1766	0.0121
Trimodal	5.226	2.291	62.55	69.22	3.6509	1.1522
Skewed Gamma	0.413	0.314	84.33	86.56	0.2249	0.1347

Posterior confidence is the mean aligned smoothed posterior probabilities assigned to the actual state. KL compares clean generating emissions with fitted emissions.

Increasing test noise: macro F1

Means over 10 paired replicates; 20,000 training and 5,000 test observations per replicate.

		Macro F1 (%); final noise SD / training SD			
Family	Model	Clean	10%	25%	50%
Gaussian	Gaussian	98.01	97.66	97.13	88.77
	GMM	97.84	97.93	96.72	88.48
Student-t	Gaussian	48.73	48.71	48.35	43.96
	GMM	95.12	94.48	89.26	72.62
Trimodal	Gaussian	56.28	56.54	56.51	56.51
	GMM	67.86	67.65	65.86	59.58
Skewed Gamma	Gaussian	82.06	81.61	80.35	74.93
	GMM	86.40	86.41	85.99	79.17

GMM retains a Student-t advantage at every tested noise level.

Switch counts expose unstable decoded regimes

Means over 10 paired replicates; 20,000 training and 5,000 test observations per replicate.

	Clean: inferred count		50%: inferred count		True count
Family	Gaussian	GMM	Gaussian	GMM	
Gaussian	27.0	27.0	23.7	23.7	29.7
Student-t	474.7	23.0	461.7	17.3	29.7
Trimodal	900.7	1043.3	903.3	1077.3	29.7
Skewed Gamma	196.3	542.7	167.0	560.7	29.7

Summary Findings

- Both model's classification accuracy decreased compared to mean regimes
- GMM still outperformed Gaussian in most metrics clean F1 on Student-t, trimodal, and Gamma data.
- Recovery and stability differ. GMM infers more switches than Gaussian on Trimodal data but has a better F1 score